DESIGN / INTERACTION / TECHNOLOGY

PORTFOLIO

Ruoyi (Iris) Zhao



01

BodyBuddy

Making body knowledge approachable



02

DribbleIQ

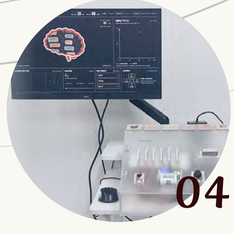
Turning movement into actionable feedback



03

Beyond The Loop

Experiencing the logic of recommendations



04

Dietary Traces

Visualising the long-term effects of diet

01
BODYBUDDY

BODYBUDDY

2025/03-2025/09

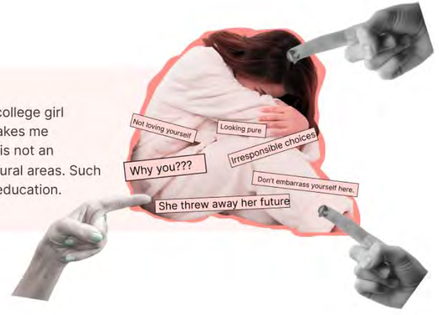
In remote rural areas of China, 62% of adolescent girls lack basic physiological cognition, and the avoidance of traditional sex education and lack of resources make girls' self-protection ability systematically lacking.
Build modular sex education aids that can be incorporated into the classroom, expose girls to sex education earlier, and teach them in a gamified format that children can better accept, teaching them to know their bodies and protect themselves at the same time.



Inspiration



On April 23, 2025, I saw a report about a college girl giving birth alone in a bathroom, which makes me shocked.After searching, I found that this is not an isolated case, it is even more common in rural areas. Such behavior may be caused by a lack of sex education.



Background

What are the reasons for the lack of sex education?

- Cultural Taboo
- Institutional flaws
- Environmental factors
- Educational methods
- Generational differences
- lack of qualified teachers



Early marriage and early childbirth are considered normal phenomena.

Lack of basic understanding of sex

Cannot face menstruation and treating it as a disease.

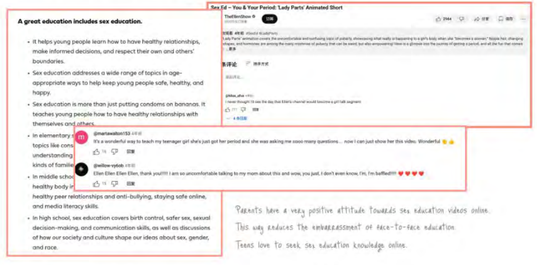
Don't know how to protect oneself when facing an infringement.

Design goals

Develop a culturally embedded, low-technology-dependent **sex education intelligent teaching aid**, and solve the problem of lack of sex education for **girls in poor areas due to lack of resources**, **cultural taboos and cognitive disconnects** through privacy-preserving design and decentralized learning mechanism.

Secondary Research

Web analysis



Cognitive stages at different ages

0-3 year *Awakening Stage*
Body parts | Perceptual learning
Basic Cleaning Knowledge

3-6 *Exploratory Stage*
Private parts | refuse touching
Gender cognition

The psychological and physiological needs for sexual knowledge among girls of different age groups

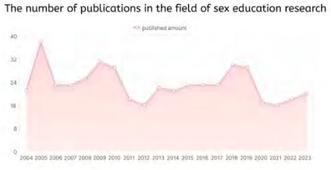
7-11 *Confusion Stage*
relationship
Physiological phenomenon
Defend boundaries

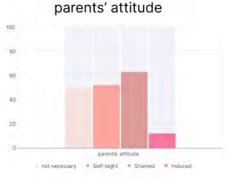
12-15 *Integration Stage*
Contraception | Intimate
Gender identity

Summary

The best age for sex education is between 10 and 15.

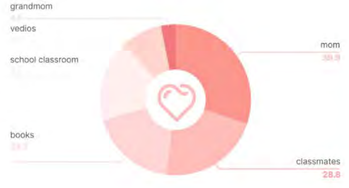
Data analysis







1.There is a lack of government attention to sex education.
2.Parents in rural areas lack knowledge about sex education.



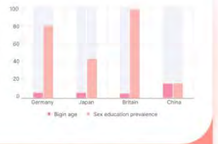
Hoping source of knowledge

33.4% Specialists | 32.4% Mom | 18% School classroom

1.Students are more eager for interesting learning methods.
2.Children's access to sex education is limited.

Sex education needs





1.Sex education spans broad knowledge and demands extended teaching time.
2.China lacks sex education compares with other countries.

Case study

Online sources



In the form of short videos, most are primarily entertainment-oriented, with compliance and educational aspects.

Picture book



The cost is low, but the static content lacks interactivity



Toy Book



Sex education apps rely heavily on the internet and are not widely popular.

Digital products



Different colors of lights indicate the alert levels of body parts.

Traditional products



Simulated fertilization process



Each age square has matching knowledge cards to learn.



Intuitively display the shape of the female reproductive organs.



Fertilization is shown as a maze, which can be used in kids' safety games.

Summary

- There are no targeted tools, and the prices are all relatively high.
- Lack of systematic sexual education hardware.
- The maintenance costs of smart devices are relatively high, making it hard to ensure long-term usage.

Literature research

Gamified learning is promising for sexuality education. It can attract the interest of young people and stimulate a variety of complex situations at the same time to stimulate students' learning potential.

In the application of educational technology, acceptability, usability and utility are the three key dimensions to consider according to the characteristics of the target group, i.e. the preferences of adolescents.

Although educational resources for sex education are available, attractive and easily accessible materials specifically tailored for young Hong Kong adolescents remain scarce. Since "Net Generation" adolescents are drawn to technology and videogaming, promoting sex education through a game-based learning approach seems promising. In a conservative culture where topics relating to sex are not openly discussed or taught by parents,[3,4] the game-based approach may also provide a risk-free, nonthreatening, and highly engaging environment in which students can learn, explore, experiment, and make decisions in stimulated situations.[8,9] Through this interactive game-based medium, learning potential is maximized.[10]

Acceptability, usability, and utility are three critical dimensions for effective use of technologies in education contexts. Acceptability refers to users' views on the value of technology (or a game) in fostering learning (e.g., whether the game content is relevant and the cost is reasonable), whereas usability relates to the user-friendliness and effectiveness of the technology (or game interface) in a given learning context. Utility, on the other hand, is relevance of the tasks in relation to the target outcomes.[11] These three dimensions can be recognized through the application of participatory design approach, a methodology emphasizing in the design process[12] as well as akeholders to design, develop, and ict.[13] The theoretical framework that

Key results of the survey supplemented with quotes from the student interviews are shown below. On average, students agreed that the game was fun and interesting (mean = 4.36). After playing the game, students believed that their ability to make smart choices in relation to love and sex improved (Section 3: mean scores ranged from 4.54 to 4.66).

In a game design called "Wise Choices", the participating children generally found the game to be fun and experienced to improve their sexual and emotional decision-making skills.

Conclusions

These findings suggest that young people are able to perceive individuals as models and that the predominant models of sexuality were found to be exploitive and manipulative. Role models convey information as well as attitudes, values and evaluative criteria, and therefore represent a broad and important source of sexual socialization. The recognition of this influence is essential if concerned adults are to utilize these in an advantageous manner.

The age of sexual initiation is getting earlier and earlier, and it is very necessary to carry out sex education early.

In high-income countries, sexual debut usually occurs during adolescence [5], though research suggests that sexual initiation is increasingly occurring at earlier ages [6]. Adolescents have to deal with the results of unhealthy sexual behaviors, including unplanned pregnancies and sexually transmitted infections [7], as well as experiences of sexual violence [8,9]. Adolescents are aware that they need more knowledge in order to enjoy healthy relationships [10], yet do not receive enough of the kind of information from parents or other formal sources that would allow them to develop a more positive, respectful experience of sexuality and sexual relationships [11].

Primary Research

Field research






Questionnaire Outline

Research Objects

- Adolescent girls (8–18 years)
- The girl's parent or guardian
- School teachers (especially class teachers, hygiene teachers, psychology teachers)
- Local Health Worker / Community Worker (optional)

Research Objects

- Examine adolescent girls' cognition/behavior regarding sexuality & physiology (including body awareness, self-protection, information access)
- Assess psychological distress & social issues from sex education gaps (e.g. shame, misconceptions; early marriage, sexual violence)
- Analyze current sex education status, communication barriers, and girls' authentic learning needs/preferences

Questionnaire

Name: Age :

Willing to learn?

	Extremely	Very	Basic
Body parts and functions			
Physical changes			
Refuse uncomfortable touch			
Encounter a dangerous solution			

Where do you learn ?

Teachers Parents Classmates Books

How well you know your body ?

Your worries when learning about sex ?

Interviews with children

Have you noticed some changes in your body?

Who do you prefer to ask when you have a physical problem?

What kind of way do you like to learn ?

How do you protect yourself when you encounter problems ?

Interviews with teachers

What is the current state of sex education in schools ?

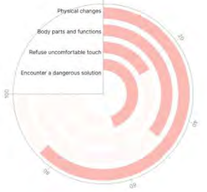
Do you think sex education is necessary ?

What kind of way to conduct sex education ?

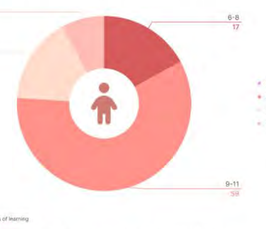
Possible problems with sex education ?

Questionnaire

1 What do you want to learn about sex?



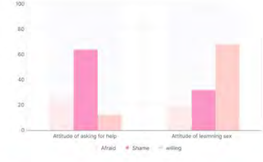
2 Age distribution of students in rural schools



3 How do you want to learn about sex?



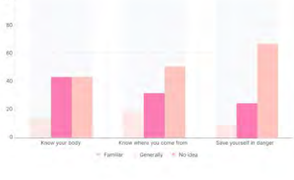
4 Attitudes towards sex education



5 When you encounter physical confusion, what do you do?



6 How well you know yourself



Interview



Q1: What do you think about the current teaching situation of sex education in schools? Are there any related courses or content?

> Our school lacks a proper sex education program currently. It's only briefly covered in biology or health classes, and even then very superficially, partly due to fear of parents' objections.

Q2: Do you think students have a need for this knowledge? What kind of behaviors or problems do they have?

> I think there's a need. Girls ask privately about their periods, and some students feel confused about interacting with the opposite sex but are too shy to ask.

Q3: What do you think would be the most suitable way to improve sex education?

> Invite specialized teachers or counselors then use picture books, animations, and interactive lessons to help kids learn naturally.

Q1 :Where do you usually learn about changes in the body? (menstruation, private parts, etc.)

> I mostly hearing from my classmates or online searches. I don't dare to ask the teacher and parents, I'm a bit shy about it.



Q2: Who do you think can talk about the topic of 'sex' ? Would you like to know more?

> I have only told a little to my good friends, and I dare not tell adults. But I actually want to know if others also encounter strange things.

Q3: What do you think would be the most suitable way to improve sex education?

> Using stories or animations, don't be too serious, and make sure the whole class of boys doesn't hear it.

Persona

Personal information

Name: Xiaohe Li
Age: 10
Occupation: student in Rural primary school

Needs

1、I don't know what to pay attention to during my menstrual period.
2、Learning through play, I don't like overly theoretical methods.
3、Eliminate body shame
4、Cleaning care, and ways to protect oneself.

Frustrations

1、Teachers and parents always skim over issues related to sex.
2、Noticing changes in my body, but don't know whom to turn to for help.
3、When discussing organs around me, I feel very ashamed.



Better understand my own body and know where I come from.



Bring sex education into the class, so that children can approach the topic with a normal mindset.

Personal information

Name: Juan Liu
Age: 29
Occupation: teacher in Rural primary school

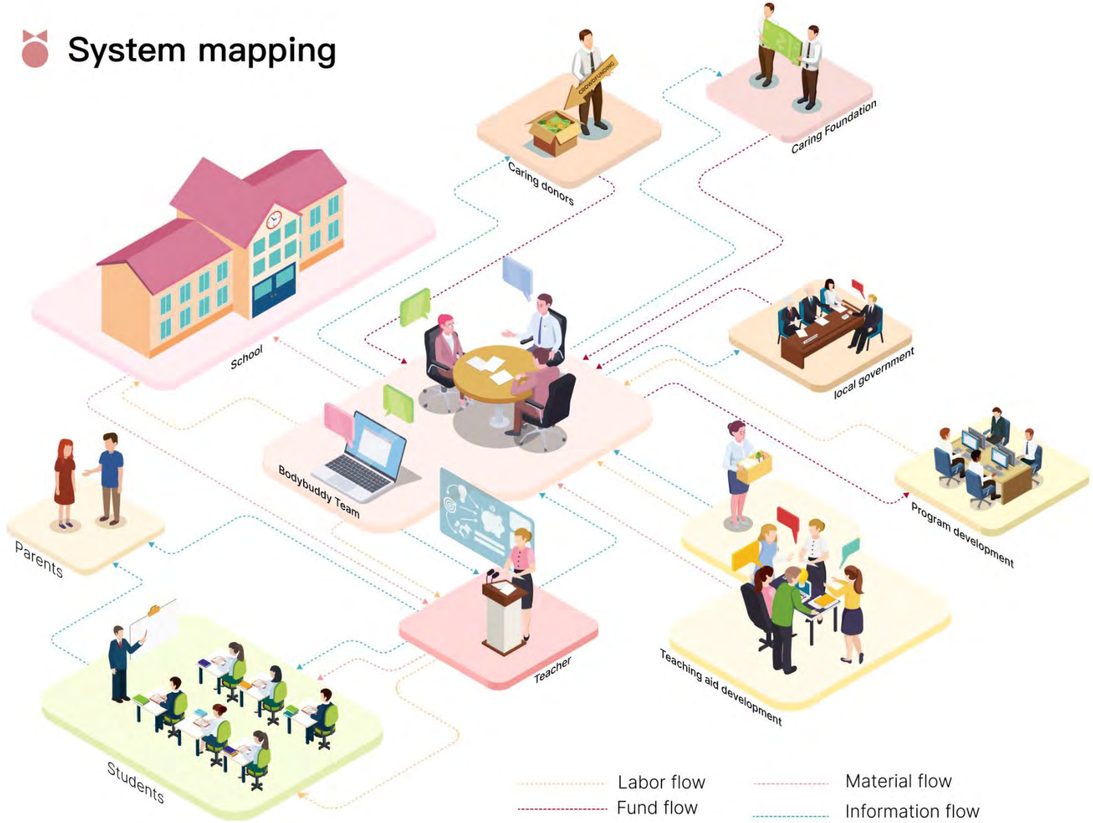
Needs

1、Make children and parents aware of the importance of sex education.
2、Systematic sex education institution
3、Simple and easy-to-understand sex education materials
4、Teach sex education to children in the classroom without criticism.

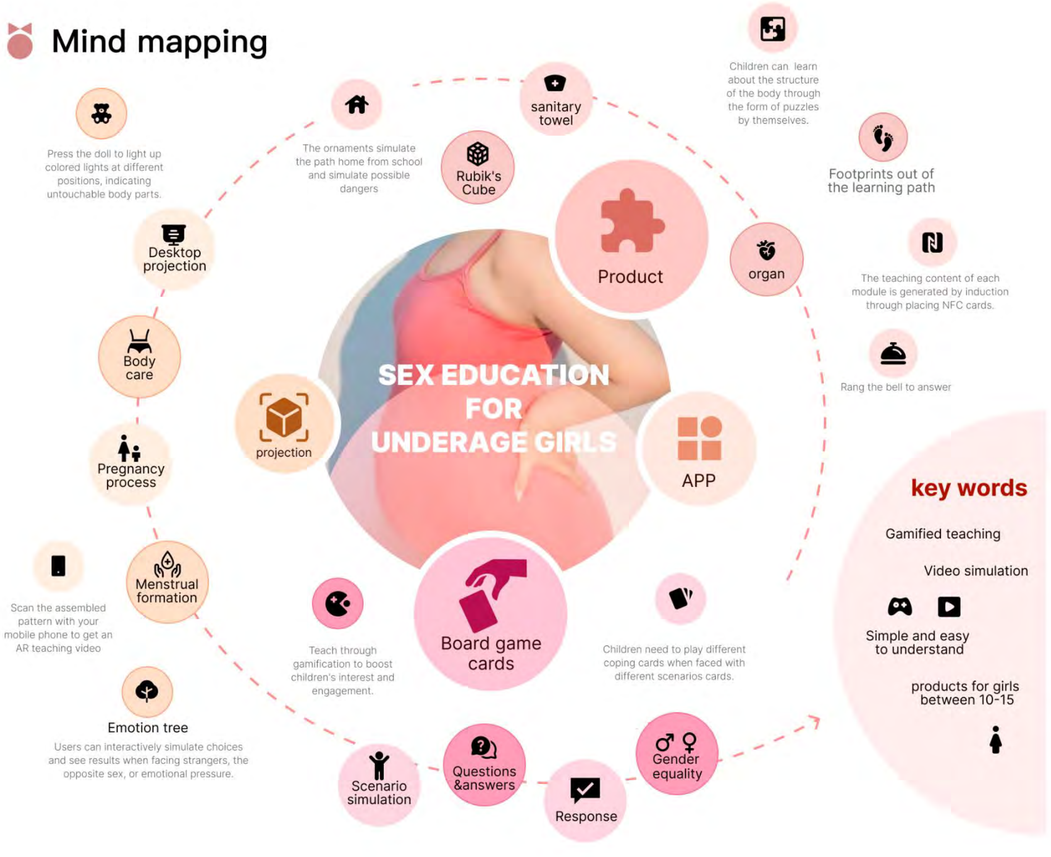
Frustrations

1、When children seek help regarding changes in their bodies, it is difficult to address with simple language.
2、Consider that it is urgent to promote sex education,but no precedents and difficult to advance.
3、Limited resources, sex education difficult to implement.
4、Parents say she brings bad influence when knowing she teaching about sex.

System mapping



Mind mapping



Final concept

Public welfare tracking and feedback system

- The App can track the delivery path and usage status of each set of teaching aids, and show donors "where the teaching aids are delivered, children's feedback and teaching site photos" to build public trust and motivation for continuous participation.



Cognitive Layering, Teacher Empowerment

- Using AI technology to provide explanations of sex education knowledge while playing games.

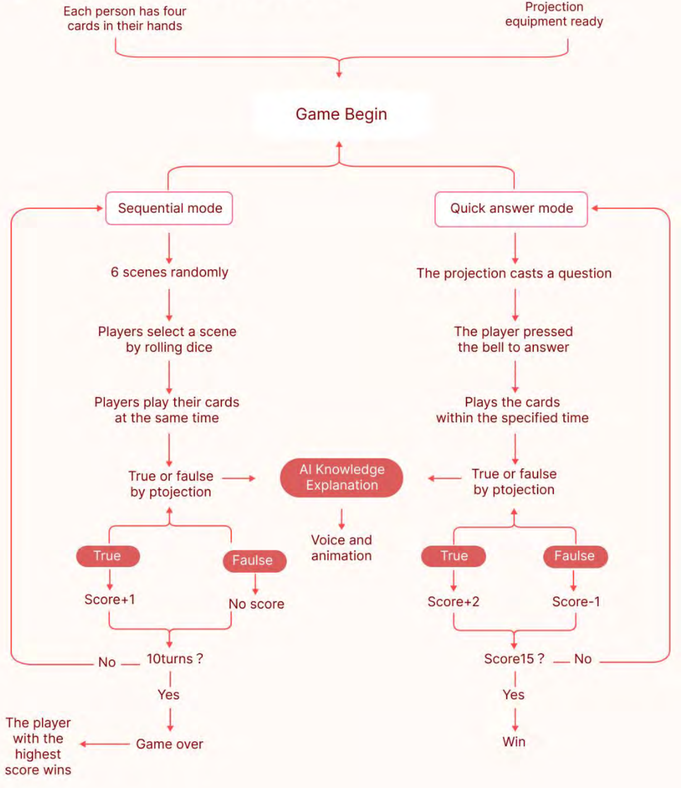
Teacher empowerment support system

- The teacher app provides built-in training manuals and videos to help non-professional educators effectively teach sex education courses.

Gamified Teaching Module

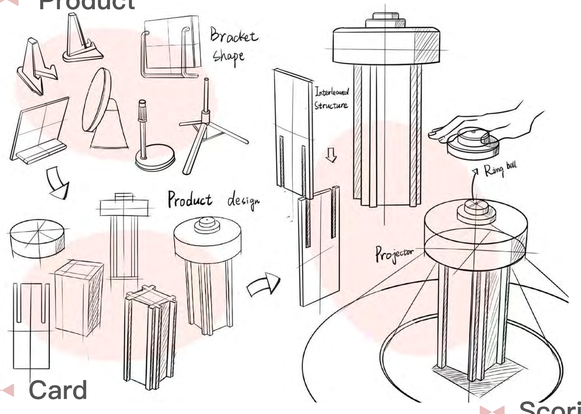
- Privacy Cards & Scenario Games: Children learn body boundaries and self-protection by simulating real situations in a "mission-based" process.
- Set up a quick answer mode to increase the fun and competitiveness of the game and enhance children's learning motivation

Game machanism



Sketches

Product



Card



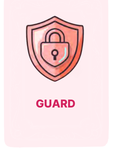
Insight Cards

Provide the correct way to deal with different situations.



Aid Cards:

Learn to get external support.



Guard Cards:

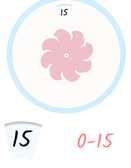
Take the initiative to avoid wind risks.



Mend Cards:

Dealing with emotions and self-identity.

Scoring wheel



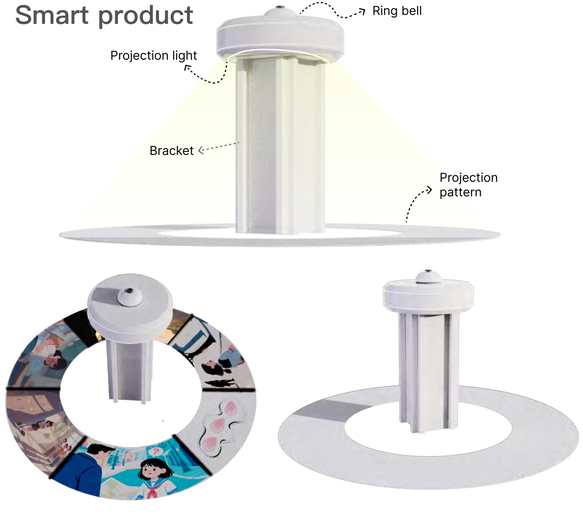
By rotating the disc above, the score marked below is revealed, with a total of 0-15 points.

Outcome

Card categories



Smart product



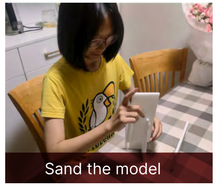
Rendering

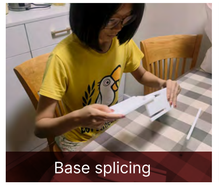


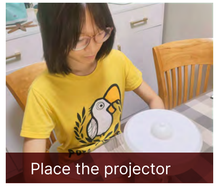
Prototype

Model making

3D printing combination







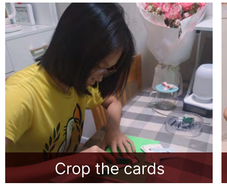
Scoreboard making



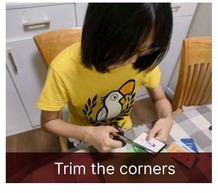




Game card making

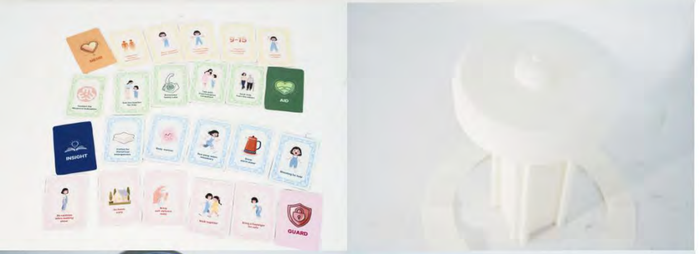






Modeling







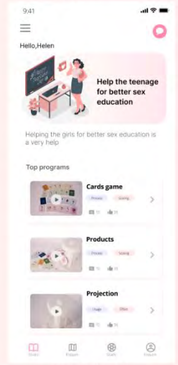
Scenario



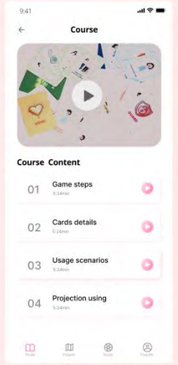
Interface design



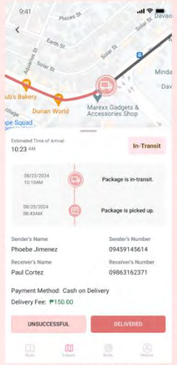
Identity Selection



Tutorials List



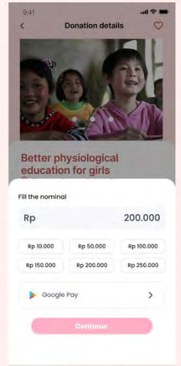
Tutorials Details



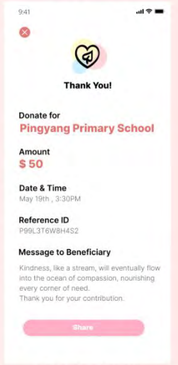
Logistics Tracking



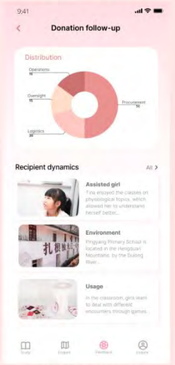
Project Details



Donation Details



Donation Results



Donation Tracking

Service flow chart

Core user

- Caring donors in society who want to help girls
- Teachers who teach physiological knowledge in rural schools
- Girls in school who are 10-15 years old and need to learn physiological knowledge

1 Donors can view project details

Front stage

- APP displays graphics/3D models of sex education teaching aids
- Displays the target amount of the project, the progress raised, and the information of the beneficiary schools

2 Donors make donations

Front stage

- Generate a donation form (select the amount payment entrance)
- Generate an electronic donation voucher

3 Buy Bodybuddy from the donated money

Front stage

- Inform donors of the progress of their purchases

Back stage

- Procurement and warehouse delivery

4 Logistics company transportation

Front stage

- Generate a donation form (select the amount payment entrance)
- Generate an electronic donation voucher

5 Donors view the progress of their donation

Front stage

- Visualization of progress timeline (procurement→ transportation→ sign-off)

Back stage

- Connect with logistics companies and track logistics progress
- Aggregate multi-source data for procurement, logistics, and receipt

6 Teaching aids arrive at the school

Front stage

- Push "Signed Received" notification
- Displays school geolocation and reception times

7 Teachers learn to use Bodybuddy

Front stage

- Provide teaching aid teaching videos/graphic guides

Back stage

- Online customer service solves problems in real time

8 Teach and use in class

9 Upload the lecture feedback map

Front stage

- Teacher side photo/video upload interface
- Donor side generates a "classroom album"

Back stage

- AI audits image compliance

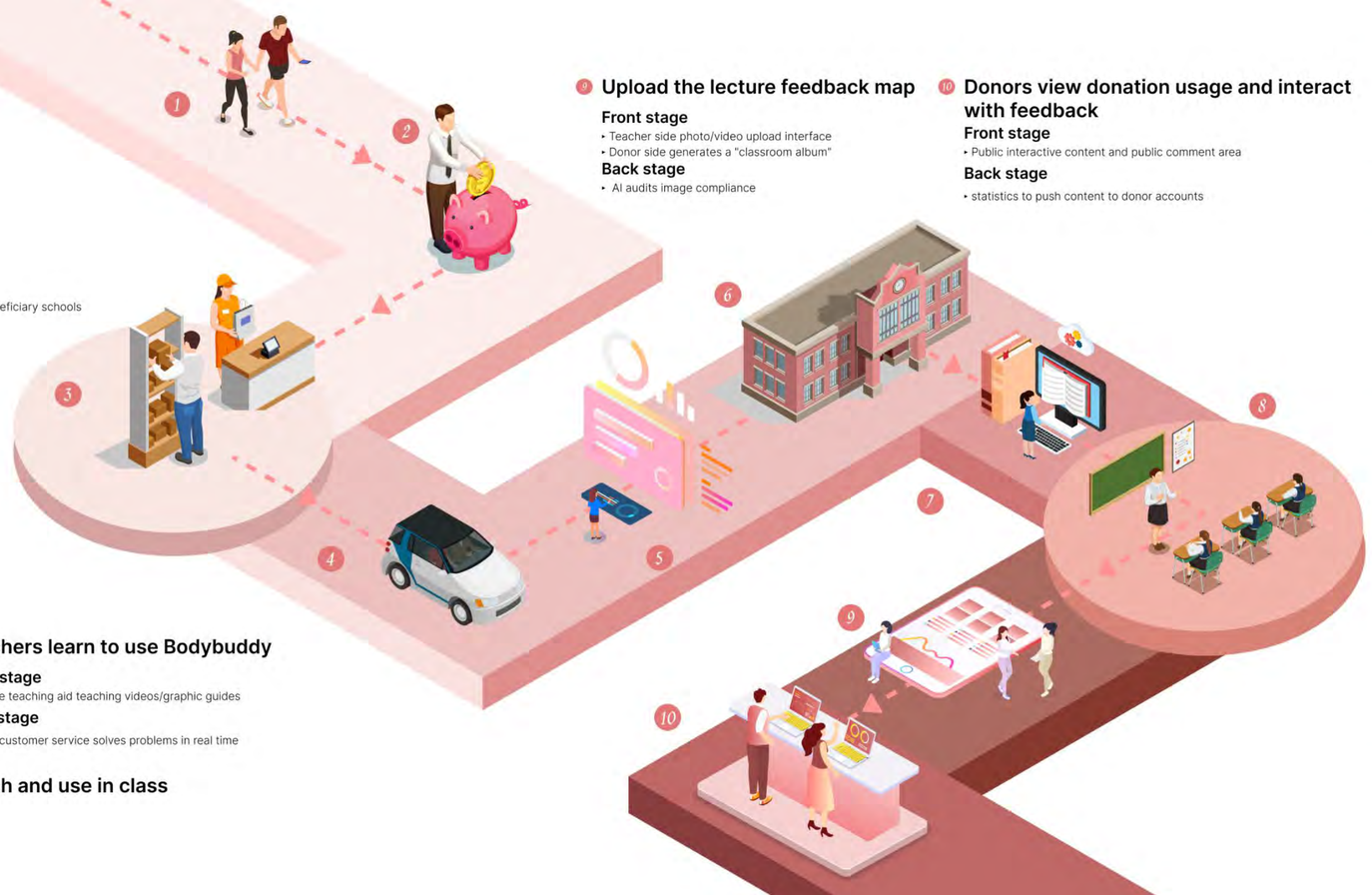
10 Donors view donation usage and interact with feedback

Front stage

- Public interactive content and public comment area

Back stage

- statistics to push content to donor accounts



Future plan

What I learnt...

Resources are limited in poor areas, so teaching aids need to have offline capabilities and non-verbal interaction functions. Based on regional adaptability, design projectors that can be used offline, combining education and entertainment. Future efforts will emphasize "design flexibility" to operate efficiently in low-resource environments.

The design of sex education aids requires interdisciplinary knowledge (like pedagogy, psychology, sociology, and public health). Through reading literature, I learnt to integrate these perspectives and transform sensitive topics into respectful, culturally appropriate visual and interactive forms.

If I had more time...

Current teaching aids focus on girls aged 10-15. Future versions may include younger children (6-9 years) and older adolescents (15-18 years). These will cover life education, gender equality, and understanding social boundaries.

Partner with schools to conduct a 3-5 year study on participants, assessing how educational tools affect their understanding of sexual assault prevention, self-view, and mental health. Publish findings to inform policy.

In the future, I will also try to reduce the production cost of teaching aids, so that more can be produced with a certain amount of donation.In addition, the donation process is made more transparent and gives donors more confidence

02 DribbleIQ

This project is an intelligent wearable system with a companion app for basketball training. Using multi-point sensing from gloves and protective gear, combined with computer vision and algorithmic analysis, it enables hand recognition, performance evaluation, and real-time feedback. The system enhances training efficiency through visual reports and guidance, helping users improve ball-handling skills and overall performance.

Interaction Design / Python / Wearable Design / APP Design

2026/01-2026/04



Team Member : Zhao Ruoyi(Me) : Product design, human key point recognition coding, large model integration and prompt engineering

Hua Li : YOLO model training and coding

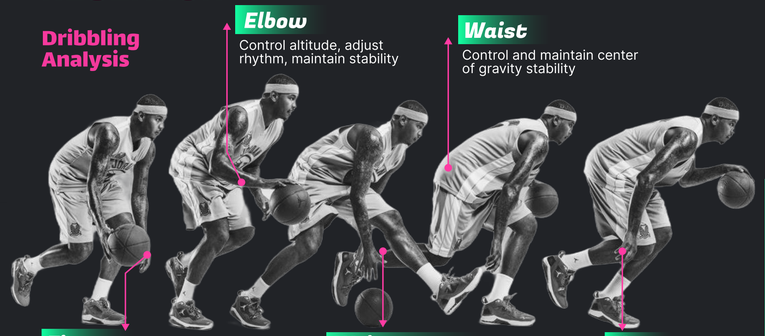
Nan Yixuan : Score Mini-Game Design and Development

BACKGROUND

Many basketball players practice individually without professional guidance or real-time feedback. During repetitive drills such as dribbling, it is often difficult for players to determine whether their movements are technically correct or whether their skills are actually improving. This lack of objective feedback can lead to repeated incorrect movements and reduces both the sense of progress and motivation during training.

RESEARCH

Dribbling Analysis



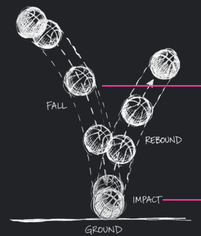
Elbow
Control altitude, adjust rhythm, maintain stability

Waist
Control and maintain center of gravity stability

Fingers
Control the direction, determine the precise landing direction of the ball.

Wrist
Control rhythm, speed, and the ball's stickiness, adjust dribbling height

Knees
Cushion and keep the upper body stable



Landing Analysis

During dribbling, the hand applies a downward force to the ball, causing it to hit the ground and rebound due to ground reaction force. The stability and height of the rebound depend on the direction, strength, and control of the applied force.

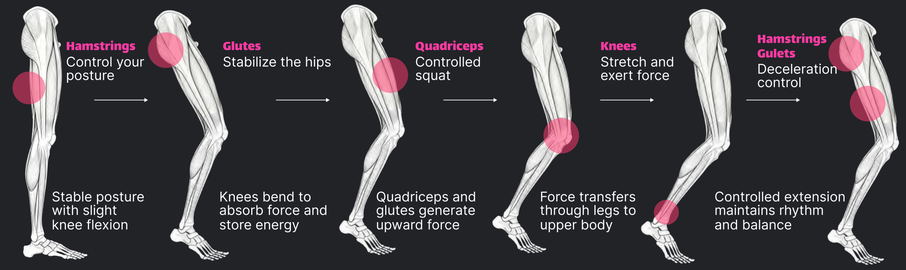
Free fall and acceleration
Under the action of gravity, it moves in uniformly accelerated straight-line motion, with gravitational potential energy continuously being converted into kinetic energy.

Energy conversion
The reaction force from the ground on the basketball reaches its peak.

Time (ms)	Net Force (N)	Kinematic Stage Notes (Do not copy to Excel!)
0	-6	Ready to push (Gravity only)
20	-15	Palm begins to apply downward force
30	-20	[Push Peak] Arm accelerates downward
80	-12	Palm thrust weakens, preparing for release
110	-6	Ball leaves hand, enters free fall
150	-6	Falling through the air
200	-6	About to hit the ground
210	0	Initial contact: Ground support force cancels gravity
215	210	Bottom of ball severely compressed, force skyrockets
218	420	[Impact Peak] Maximum deformation of the ball
221	210	Ball begins restitution (springing back)
225	0	Deformation mostly recovered, about to leave ground
228	-6	Ball completely airborne, gravity only again
300	-6	Rebounding upward through the air
400	-6	Nearing the apex, hand is about to make contact
410	-15	Hand contacts ball, begins applying downward cushion
430	-38	[Cushion Peak] Wrist flexes to absorb upward momentum
460	-10	Upward velocity approaches zero
500	-6	Ball rests in hand, completing one dribble cycle

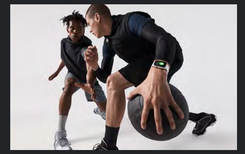
Schmidt, M., Jaitner, T. and Nolte, K., 2021. Kinematic analysis of the basketball dribbling using wearable inertial sensors. Sensors, 21(11), p.3854.

Lower Limb Force Analysis (example)



Collecting Methods Of Dynamic Data

MOST Wearable Devices

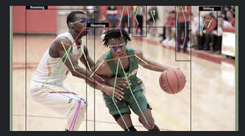




The penetration rate of smart wearable devices continues to grow

LESS Camera / Pose Estimation





Pose recognition has high accuracy but is easily affected by the environment.

Visual capture accounts for a relatively high proportion in indoor fitness and AI training.

The industry trend is shifting from a single visual solution to a '**visual + sensor**' integrated model.

Dribbling Skills Improving Observation



FAST OVERTOPS

30 pounds each hand



WIDE CROSSOVERS

30 Flour Touches



POCKET PULLS

30 pounds each hand

INTERVIEW



YiMing Wang

- 20 years old
- Basketball major student at physical school



TianYang Feng

- 21 years old
- School basketball team player

How do you practice dribbling ?	In each class, we take turns practicing specific dribbling moves, from stationary to on the move, 10 minutes each.	I usually don't practice dribbling specifically, except before a game, or when imitating the dribbling moves of basketball stars.
What difficulties did you meet?	The exercises in class are relatively basic, and I want to learn more professional technical moves, footwork, and dribbling skills.	I don't really know how to practice dribbling, I just keep repeating a few specific moves, and the coach doesn't specifically correct my movements either.
What help do you need?	I want something can keep my current skills sharp, maintain my touch in practice and learn new moves and techniques to improve myself.	I hope there is something that can tell me if my movements are correct, and can help me figure out what to practice to improve my skills.
What kind of feedback do you hope to receive?	I hope to receive professional training data analysis, as well as whether my training has been effective, and what progress I have made.	Correct my movements, steadily improve my skill level, and be able to perform normally on the field
Contrast	• Structured and consistent practice routines • Strong understanding of techniques and details • Focus on refining skills and improving performance • Require precise, data-driven analysis	• Irregular and unstructured practice • Lack of knowledge about correct techniques • Want to improve but lack clear direction • Need simple guidance and real-time correction

COMPETITIVE ANALYISIS

投篮实验室Shotlab

Committed to empowering all basketball scenarios through AI smart hardware, improving the entertainment, training, and competition experience of all basketball groups starting from shooting.



Strength

- Can comprehensively analyze shooting trajectory, body posture, shooting angle, and other data.
- With AI and video replay, it feels very professional.
- Users can compete online or offline, combining professionalism with fun.



Weakness

- Lacks real-time feedback; results analysis can only be seen after training.
- Highly dependent on the environment, users need to train at specific venues.
- Lacking teaching modules, users do not know how to train the correct movements.
- Did not provide diverse shooting action guidance, only basic analysis.
- The UI design lacks aesthetic appeal and is too rigid.



Threats

- Increasing competition from similar AI training products.
- High equipment and venue dependency limits user expansion.
- User attrition due to low engagement and weak feedback.
- Limited scalability due to high costs of opening new locations restricts geographic expansion.

Opportunity

- Vision-based systems struggle with occlusion and hand-level precision.
- Introduce wearable sensors to capture fine-grained motion data.
- Enable more accurate and reliable basketball training feedback.

DESIGN STRATEGY

Design Question

Professional trainees find it difficult to receive one-on-one guidance.



Amateur enthusiasts lack access to professional training, or the training costs are expensive.

While standardizing correct movements, protect athletes from injuries.

Design Target and Goals



Active basketball players or potential athletes receiving professional training

- **High-precision data**
- **Technical Details Analysis**
- **Improve motion capture accuracy**



Professional user

- **Real-time feedback**
- **Enhance training motivation**
- **Establish correct movements**

PERSONA



John

Professional user

Second-year basketball specialiststudent at Physical School

Point guard

Training Goals

Improve skills and become a main player on the school team.

Demand

- **Clearly quantify the dribble stability**
- **Suitable training plans**
- **Visible progress**
- **Real-time feedback during training**

Pain Point

- Existing training is too basic to support advanced skill refinement.
- Difficult to identify subtle mistakes in technique and movement.
- Hard to measure training effectiveness and long-term progress.



Simon

Amateur user

University varsity team

Point guard

Training Goals

Maintain the feel and be able to perform normally on the court.

Demand

- **Clear training guidance**
- **Real-time feedback**
- **Simple performance insights**

Pain Point

- No clear plan or system for practicing dribbling.
- Do not know what to practice or how to improve.
- Mistakes go unnoticed and become habitual.

CONCEPT

My design includes two parts.

The first segment consists of wearable devices that employ a fusion of multi-sensor arrays and nylon fabric materials to implement data monitoring, recording, and vibration feedback functionalities.





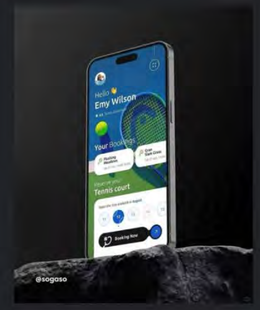
The second component entails the development of a unified application, converging core modules such as systematic reporting, training analytics, video motion capture, video tutoring, and social community to orchestrate a seamless end-to-end user experience.

TRAINING
Multiple training modes support different stages, helping users advance from beginners to advanced levels, providing professional movement guidance.

ANALYSIS
Track training data and generate visual reports with key metrics and progress trends.

PRODUCT
Offer equipment and virtual content, with smart recommendations based on training data.

RANKING
Leaderboards and achievements showcase performance and motivate continuous training.

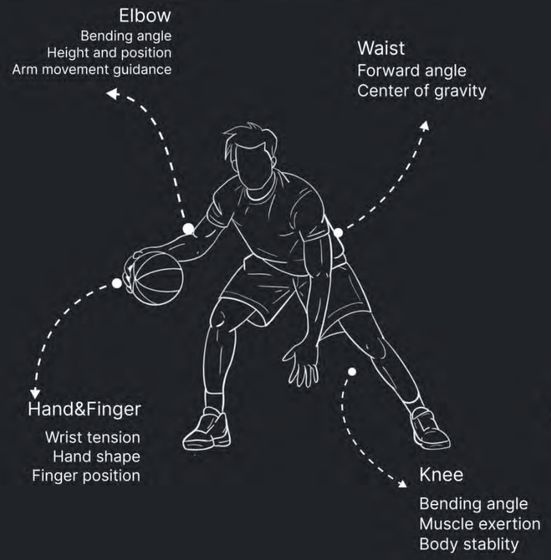


WEARABLE PART SKETCH

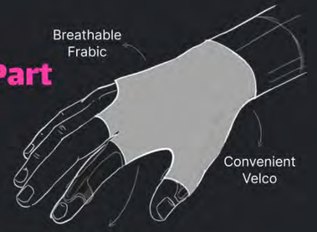
Wearing position

I choose the wearing location based on the function of each body part during dribbling.

- **Hands** → Touch and ball control
- **Elbow** → Action Execution
- **Knee** → Lower limb exertion
- **Waist** → Core stability control



Hand Part





Embedded Sensors

The sensor is embedded inside, without affecting wearing or training

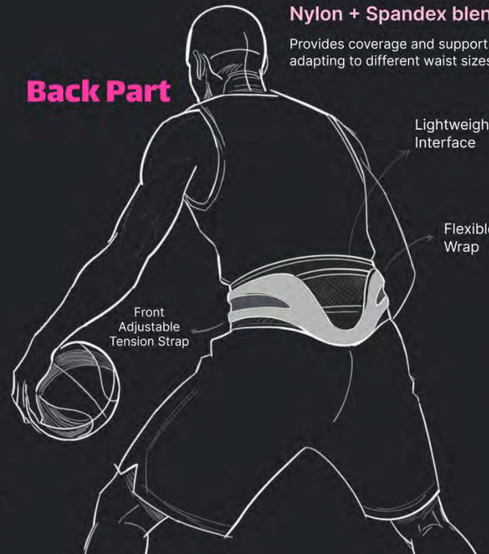


Pressure sensor

FSR402 Resistive Linear Pressure Sensor

- Record ball touch situations
- Correct dribbling hand position

Back Part



Elastic Band
Nylon + Spandex blend

Provides coverage and support while adapting to different waist sizes

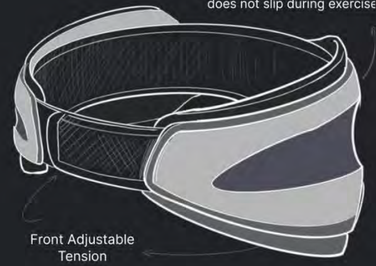
Section Detail

Fits the curve of the waist, ensuring comfort



Dynamic Mobility

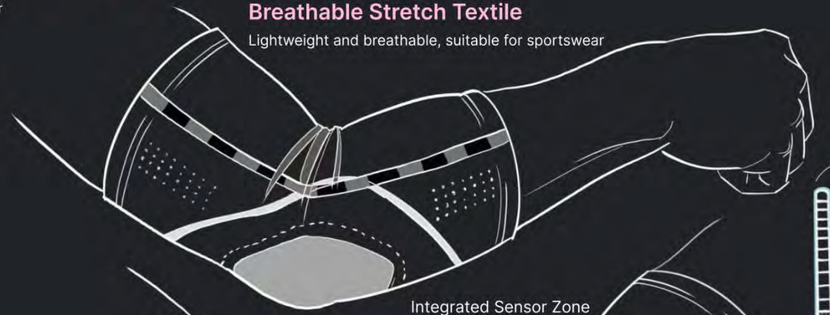
Can be customized to ensur
does not slip during exercise



Elobw Part

Breathable Stretch Textile

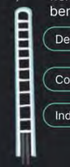
Lightweight and breathable, suitable for sportswear



Bend Sensor

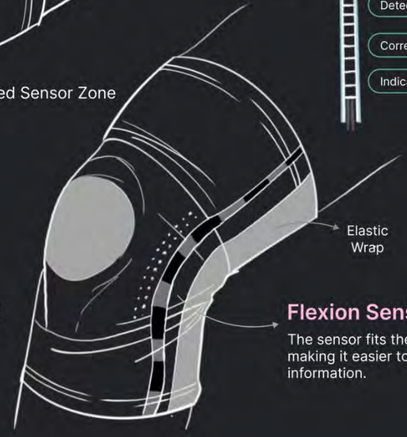
flex 2.2 unidirectional bending sensor

- Detect joint angle changes
- Correct arm position
- Indicate lower limb power status



Integrated Sensor Zone





Flexion Sensor Array

The sensor fits the curves of the leg, making it easier to recognize information.

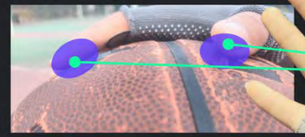
PRODUCT TESTING

HANDS:
Semi-enclosed glove style, pressure sensors at the second knuckle.



PRESSURE SENSOR
Detect dribbling force, touch frequency, and ball control stability.

IMPROVEMENT
Move the sensor to the second joint of the finger, which has less impact on dribbling.





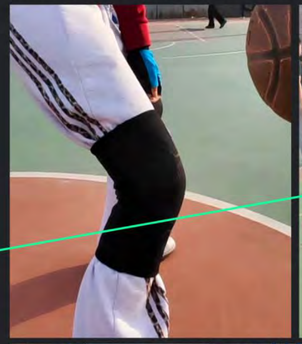
WAIST:
Exercise waist support style, sensors on both sides.



BEND SENSOR
Both sides of the waist Identify whether the sides of the waist are exerting force when bending.

Detects leaning forward/backward of the body.
Sensing overall center of gravity changes.

KNEES & ELBOWS:
Knee and elbow guards style, side bending sensors, sense bending angles.









TEST RESULTS
Has a strong sense of being wrapped, does not affect the normal bending of arms and legs

Built-in sensor, does not affect movement.

IMPROVEMENTS
Switching to a Velcro closure might make it easier to put on and take off

3D MODEL AND RENDER













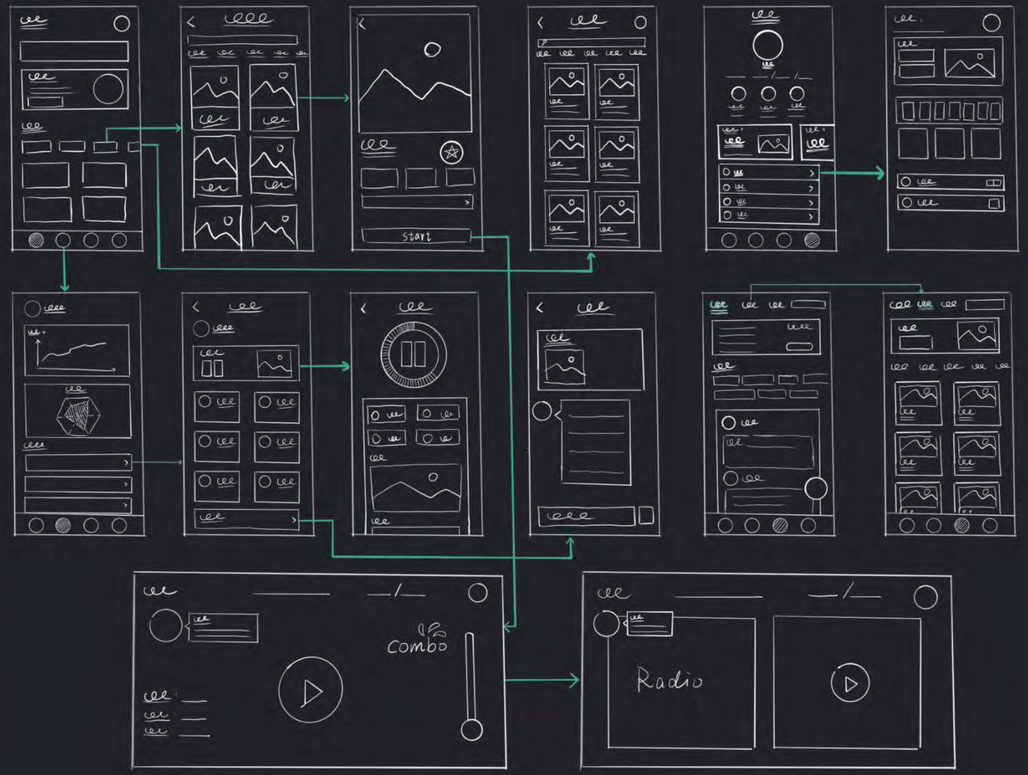


APP DESIGN

Information Architecture



WireFrame



Viusal Component

Emotional Version



Sports | Professional | Enthusiasm

Color Specification

#5CAC8C

#FCFF6E

LOGO

DribbleIQ

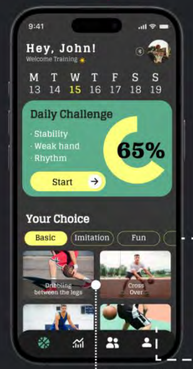
DribbleIQ

Font

Glegoo bold

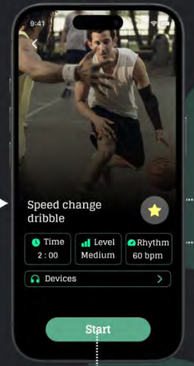
Glegoo Regular

UI Design









See more actions

Different focus in each part

Four Training Mode Available

- Basic
- Player Imitation
- Fun
- Combat

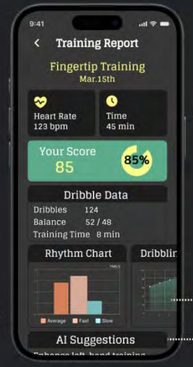
Multiple Actions Available

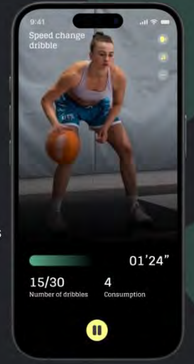
Favorite actions

Personalized settings

- Training Duration
- Difficulty Level
- Other Parameters







Report Page

You can view your progress curve here and filter by time and skill dimensions.

Understand your strengths and weaknesses through a radar chart.

View past training reports

Chart

Visualize dribbling stability, rhythm, and other aspects.

Personalized section

- Habit Analysis
- Cause judgment
- Training Recommendations

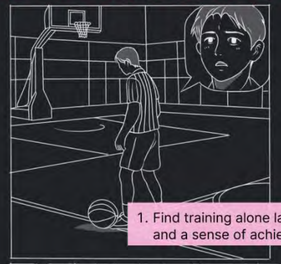
Analysis content

- Action Association
- Point of progress
- Growth Record

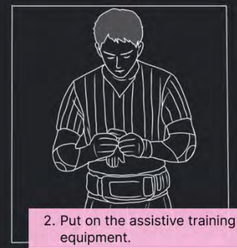
DribbleIQ



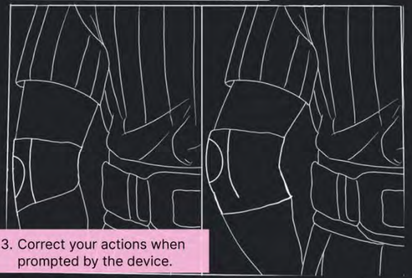
STORY BOARD



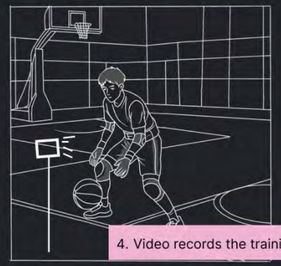
1. Find training alone lacks motivation and a sense of achievement.



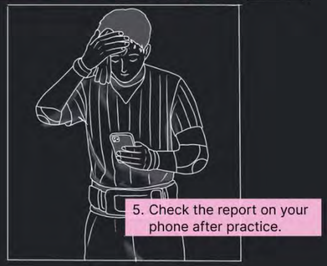
2. Put on the assistive training equipment.



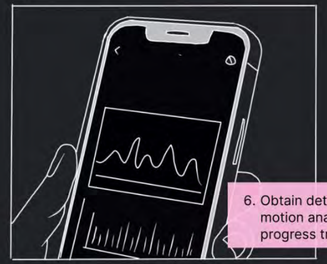
3. Correct your actions when prompted by the device.



4. Video records the training process.



5. Check the report on your phone after practice.



6. Obtain detailed motion analysis and progress trend data.

PYTHON TECHNIQUE



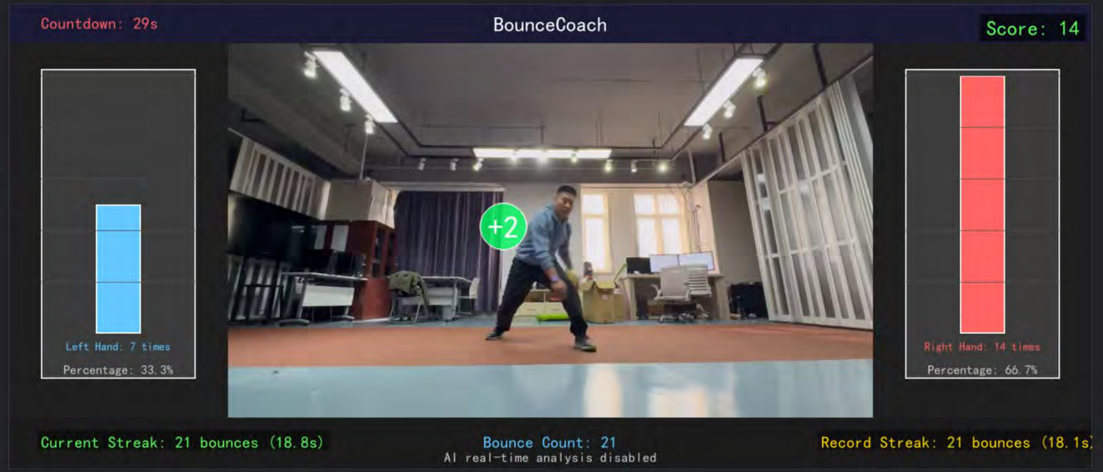
Demo Video :https://www.youtube.com/watch?v=qIHNctSs5iE

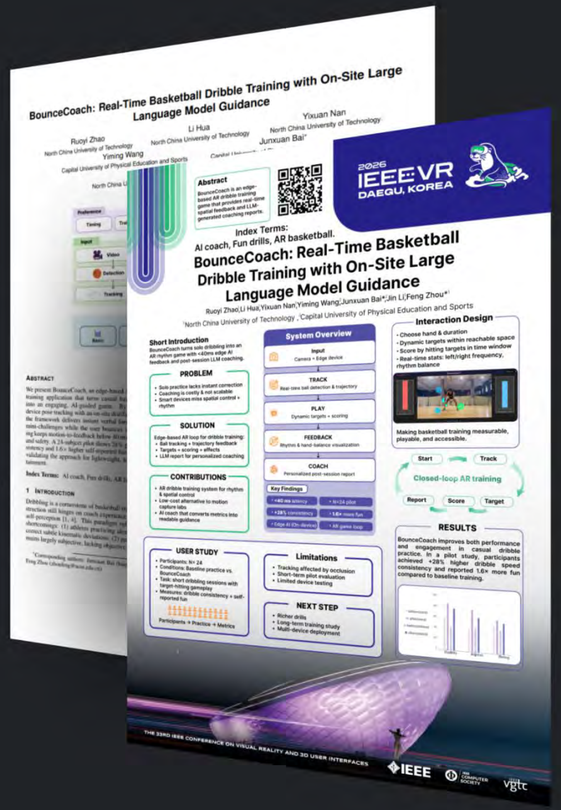
Technical development achievements

This design project is based on the basketball training system my team and I developed, combining real-time computer vision, edge computing, and AI-driven interactive design.

The study was accepted by a workshop (CCF-A) at IEEE VR 2026, showing that the project not only contributes to design exploration but also has recognized technical research value.

Paper link:BounceCoach: Real-Time Basketball Dribble Training with On-Site Large Language Model Guidance





03

Beyond The Loop

VR Digital Media Serious Game

2026/05-2026/07

Driven by personalized recommendation algorithms, we are gradually getting trapped in information cocoons. This project designs a VR serious game themed around 'algorithm manipulation.' Through immersive interactions, it turns abstract data algorithm recommendation mechanisms into a perceptible spatial interactive gaming experience, guiding users to understand the relationship between recommendation algorithms, commercial interests, and information bias, while exploring ways to break free from information cocoons.



Inspiration

As AI and algorithms continue to permeate daily life, people are increasingly accustomed to relying on systems to make choices for them: what content to scroll through, which viewpoints to follow, and even how to think.

I realize that we think we are acquiring information actively, but many times we are actually just continuously consuming the content that the system wants us to see.

Being in such an environment for a long time, people gradually fall into an information cocoon and lose the ability to explore the unknown and make independent judgments.



Data

OpenAI

ChatGPT handles over **2.5 billion** prompts each day

American employees using AI: 21% → 40%







Desk research

Why we dependent on algorithms increasingly?



As users increasingly rely on algorithms to filter overwhelming information, platforms gain more behavioral data to predict, optimize, and shape user preferences.

How do platform predict?



Watch Time
The longer a user watches, the more the system assumes interest, reinforcing similar recommendations over time.

Repeated Viewing
Rewatching, pausing, or slowing down on certain content signals emotional engagement.



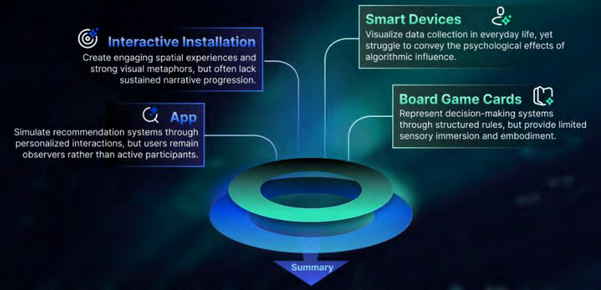


User Profiling
Platforms combine browsing habits, interactions, location, and social connections to build detailed user profiles.



Click-Through Rate
Every click provides direct feedback, helping platforms learn what captures users' attention and interests.

Output Format Analysis



VR transforms algorithmic influence into an embodied experience, allowing users to physically feel their world becoming narrower.

Design goal

01 **Turn invisible manipulation into a visible experience, and concretize the process.**

02 **Making users aware of the convenience brought by algorithms also gradually reduces the behavior of actively exploring and thinking.**

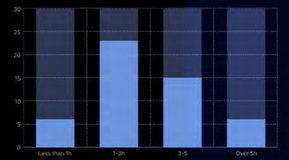
03 **Awaken people's subjectivity, maintain awareness of one's own actions, and avoid completely letting AI take the initiative.**

04 **Using a gamified approach, VR manipulates algorithms to 'perform for you,' allowing you to re-experience and reflect on your relationship with algorithms.**

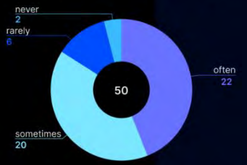
User research

Questionnaire

Q1. How much time do you spend on short video platforms each day?



Q2. Do you feel that recommendation algorithms understand your interests increasingly well?



84% of people feel that algorithms understand them better and better.

Q3.What is your attitude regarding the matter of 'the platform understanding you better and better' ?



More than half of participants expressed mixed feelings toward personalized recommendations, appreciating their convenience while remaining concerned about their influence.

Q4. Express your level of agreement with the following statements

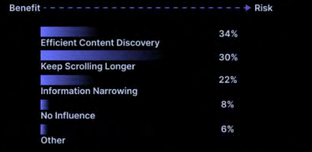


Q5. Are you aware of the formation of information cocoons and taking action?



Most participants were aware of algorithmic influence, but few took action to resist it. Awareness alone did not lead to behavioral change.

Q6.Which aspect do you think the recommendation algorithm affects you more?



Participants viewed algorithmic recommendations as both beneficial and restrictive. While 34% valued their efficiency, 52% associated them with excessive engagement or reduced information diversity.

Interview

Flora
22 years old
YouTube Veteran User



Q1: Have you ever thought about actively searching for other content? Or do you mostly just watch the recommended content? Why?

> I mostly rely on recommendations and only search for travel, shopping, or specific information. They match my interests and are easy to scroll, even if they keep me in a content bubble.

Q2: What do you think is the main reason you are aware of the existence of the information cocoon but do not take action?

> Exploring different content takes extra effort, while watching videos is for relaxing. The problem isn't serious enough for me to change my habits.

Q3: You said you are getting more understanding of algorithms and have some concerns. What exactly are you worried about?

> What worries me most is that it knows what attracts me better than I do. I'm also concerned that I'll encounter fewer new ideas, making my choices feel less free.

Persona

Pain Points

- Not realizing that sources of information are becoming increasingly uniform
- Lack of proactive exploration and independent thinking
- Unaware of the formation of an information cocoon

Needs

- Understand how recommendation algorithms shape information exposure
- Recognize the existence of filter bubbles
- Reflect on personal media consumption habits
- Develop awareness of diverse perspectives





Pain Points

- Knows they are being influenced but struggles to change.
- Lacks practical ways to explore beyond recommendations.
- Becomes less tolerant of unfamiliar or conflicting viewpoints.

Needs

- Understand the connection between behavior and filter bubbles
- Learn practical ways to break out of information bubbles.
- Engage with diverse viewpoints beyond personalized recommendations.

Case study

We Become What We Behold



Independent game

A short and sharp satirical game created by Nicky Case, themed around the media, demonstrating the concept of vicious cycles in under 5 minutes.



Game Theme

This game explores the feedback mechanism of the vicious cycle between the media and its audience through dark humor and biting satire. It analyzes, from an impartial perspective, how sensational news exacerbates division and tribalism.



Project Introduction

This VR project explores how personalized content creates filter bubbles and limits diverse perspectives.

Player experience

Players take on the role of a bee exploring a virtual multi-species garden, where each flower type represents a different perspective. Pollinating certain flowers causes them to multiply while other flowers gradually disappear.





Revelation

By transforming this abstract concept into an enjoyable and immersive experience, the project makes the concept of information cocoons more relatable to everyday life and easier to understand.

Bubbleverse





Project format

A spatial installation that transforms the idea of a filter bubble into an immersive environment. Users are surrounded by familiar information, highlighting how digital platforms can shape and narrow perception.

Inspiration

Information bubbles can be represented not only through screens and interfaces, but also through immersive environments that make invisible information structures physically perceptible.

VR Analogy for the Filter Bubble

Summary

2D Algorithm Visualization	Lack of Immersive	Environmental Metaphors	Repetitive Content
Most projects represent recommendation systems through 2D interfaces and visualizations.	Few projects fully explore algorithmic influence through immersive VR and embodied interaction.	Some projects use environmental changes and shrinking worlds as metaphors for the formation of filter bubbles.	Projects use repetitive recommendations and similar content to reinforce user interests.

Mind mapping

Brainstorm different narrative forms for the story.

Commercial Street

- Advertising
- User Browse
- Preference Learning
- Personalized Feed
- Information Narrowing



Personalization gradually replaces diversity, shaping what users notice and choose.

Data Flow Space

- User Behavior
- Data Collection
- Commercial Weighting
- Recommendation Output



Hidden recommendation logic reveals how commercial interests influence content visibility.



Each choice reshapes the knowledge space, making familiar perspectives increasingly dominant.

Librabry

- Choose Book
- Shelf Reorganizes
- Similar Topics Expand
- Alternative Views Fade



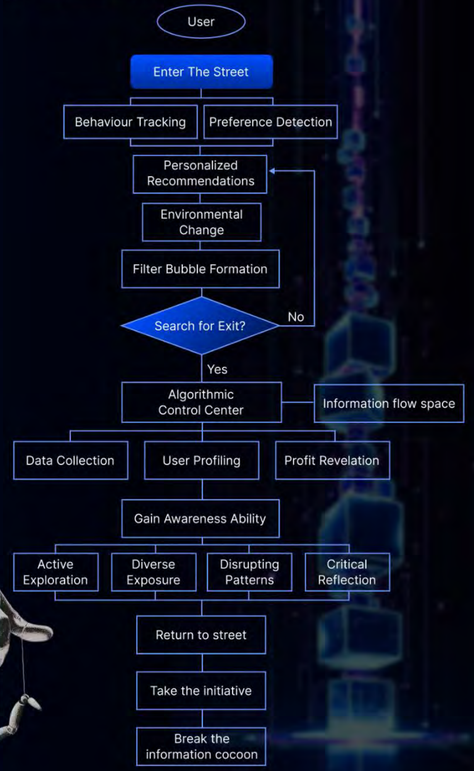
Repeated recommendations gradually become invisible walls.

Maze

- Explore
- Repeated Guidance
- Circular Paths
- No Exit



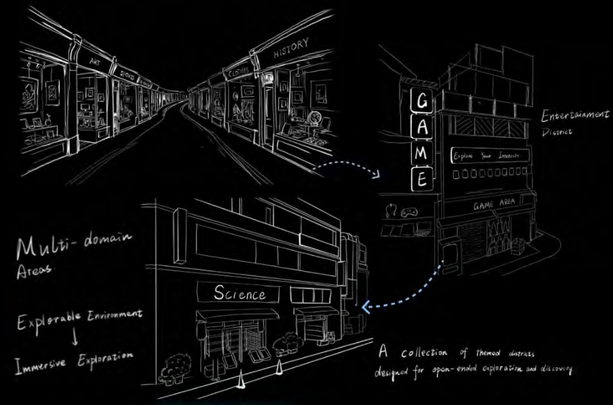
Game flow



Design process

Moodboard & Sketches





Game map

The game world is composed of multiple knowledge districts, each representing a distinct information ecosystem. Players are free to explore different topics while gradually developing unique patterns of interest through interaction.



Data Flow Space



After exploration, players enter a hidden algorithmic space where recommendation mechanisms become visible, revealing how behavioral data shapes user preferences and ultimately contributes to filter bubbles.

Reflection



By exploring different personas and diverse information domains, players are encouraged to break the filter bubble and take control of their information choices.

VR interaction technical flow

3D model- blender&unity



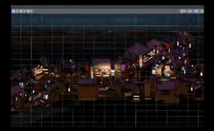
Explore Information

Users freely navigate across different information districts.



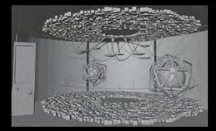
Capture Behaviors

Every interaction contributes to an evolving behavioral profile.



Adaptive Environment

The virtual environment dynamically adapts to inferred interests.



Reveal Algorithm Logic

Recommendation mechanisms and commercial influences become visible through spatial metaphors.

User interface display

Game Start



Exploring different fields



Preference Prediction



Recommendation rating



What's the next step?



System recorded data



Delete the tags you don't like



Final Revelation

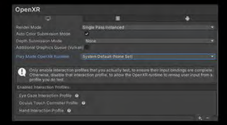


XR Environment Configuration

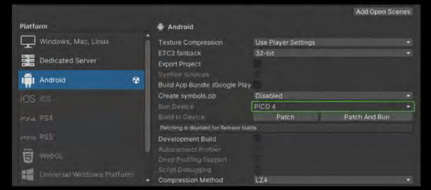
- Unity Scene
- XR Origin+Interaction Toolkit
- Andriod Build(APK)
- PICO4 Headset
- Real-time VR Experience

The project was developed using Unity XR Plug-in Management to enable VR device compatibility. OpenXR/ PICO XR providers were configured to establish communication between Unity and the headset.



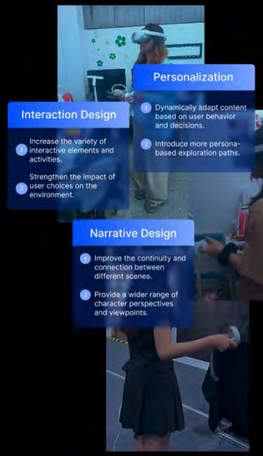


XR Interaction Toolkit was integrated to handle VR interactions, including controller input, ray-based interaction, and object manipulation.



User testing feedback

Awareness	Recognized algorithm influence	Reflected on media habits	Increased awareness
Engagement	Strong immersion	Smooth experience	Some guidance unclear
Understanding	Understood filter bubble formation	Learned recommendation logic	Understood algorithmic manipulation
Gameplay settings	Easy to understand	Interactions are engaging	Some interactions feel repetitive



Summary

Advantages

- Users showed increased awareness of algorithmic influence.
- The VR experience was engaging and immersive.
- Most users understood the concept of information cocoons.

Disadvantages

- Some interactions felt repetitive.
- Certain concepts required clearer explanations.
- Users expected more personalized and diverse content.

Reflection

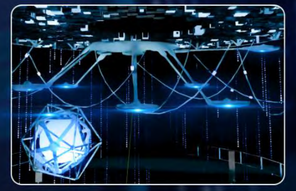
User testing suggests that the project successfully raises awareness and promotes reflection on information consumption habits. However, future iterations should focus on improving interaction diversity, narrative clarity, and personalization to create a more impactful learning experience.

Outcome & Description



A continuous stream of data flows from the outside into the balance

The size of the data flow can cause the balance to tilt differently, and those with a low tilt angle will come into the user's view.

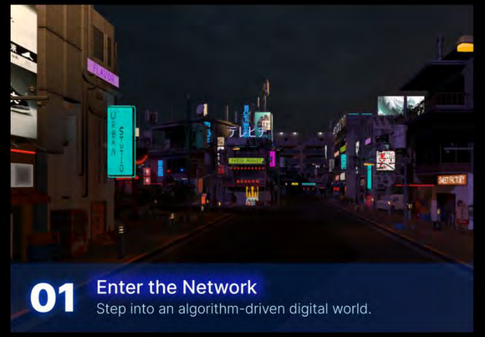




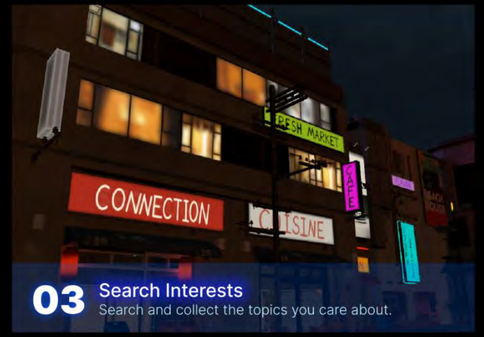
Records user preference information in the space and makes recommendations based on past choices

This space reveals the information order shaped by algorithms. Each balance represents an information domain, where the flow of interests determines content visibility. Content driven by commercial value enters users' view, while others gradually fade away. Behind seemingly free choices lies a system shaped by algorithms and interests.

Game scenario



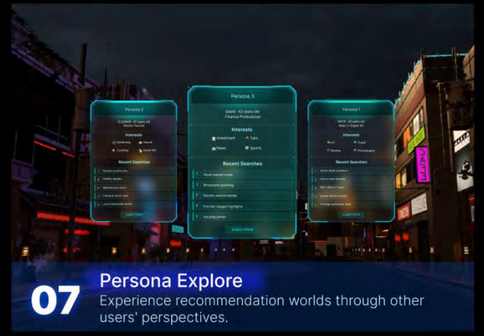














Tracking service

After the experience, users can track their information habits and turn algorithm awareness into long-term behavioral changes through sharing, check-ins, and data feedback.



Future Work

- Refine gesture recognition and interaction logic for more intuitive VR exploration.
- Explore new interactions to enhance users' understanding of algorithmic influence.
- Optimize interaction flows and content presentation based on user feedback.
- Improve visual effects and data visualization for a more immersive experience.

04
Dietary Traces

Data Translation · Interactive Installation · Experiential Learning · Technical Implementation

2026/07-2026/09

This project is an interactive installation exploring the relationship between diet and aging. Through physical food cards and data visualization, it translates daily nutrient intake into dynamic changes in human body systems. By selecting food inputs and adjusting their quantities, users can observe the long-term effects of dietary factors such as sugar, salt, and fat on bodily conditions, fostering a more intuitive understanding of the relationship between dietary habits, health, and aging.



Background

Young adults increasingly care about their health, yet everyday dietary choices are often guided by convenience, pleasure and routine. While the effects of eating can be felt immediately, its long-term consequences on the body are much harder to recognize.

Why do everyday choices feel disconnected from future health?



Individual Choice

A single meal rarely provides visible feedback about its long-term impact on the body.



Repeated habit

Unhealthy choices become part of daily routines due to taste,convenience and environmental cues.



Long-term Exposure

Over time, these habits can increase the risk of obesity, chronic diseases and reduced quality of life.

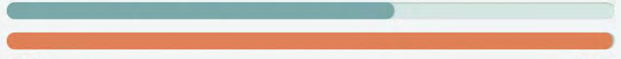
Early Health Risks

Metabolic health concerns can emerge long before old age.
A global study found rising type 2 diabetes incidence in ages 15–39.

BMJ · 2022 / 15–39 YEARS / GLOBAL ESTIMATES

1990 · INCIDENCE / 100K	2019 · INCIDENCE / 100K
117.22	183.36

Age-standardized new cases per 100,000 people



KEY FINDING

Metabolic health risks can appear in young adults, not only in old age. This trend is not caused by diet alone.

Secondary Research

Quantifying Everyday Intake

Familiar foods contain measurable nutrients, but their amounts are hard to picture. Comparing a food example with a daily reference makes those numbers easier to interpret.

01 / FOOD TO NUTRIENT QUANTITY



COLA · 330 mL

35 g

Sugar per 330 mL can
Cola · product serving example

COMPARED WITH DAILY FREE-SUGAR REFERENCE

70% 35 g / 50 g at 2,000 kcal/day

WHO: free sugars < 10% of daily energy

02 / OTHER DAILY INTAKE REFERENCES

01	SALT Adults · salt, not sodium	< 5 g/day
02	DIETARY FIBRE Ages 10+ · naturally occurring	≥ 25 g/day

Sources: Coca-Cola Original Taste (330 mL); WHO Healthy Diet.

COLLEGE STUDENT DIETARY INTAKE
762 US students · Observed average daily fibre

15.46 g/day

WHO REFERENCE ≥ 25 g/day

Sample average ≈62% of 25 g reference; not representative of all youth.

The Invisible Progression

CHANGES BEFORE SYMPTOMS

Long-term diet-related health risks may develop gradually. Some metabolic and cardiovascular changes can occur without noticeable symptoms, creating a gap between body and perception.

WHAT WE FEEL
Subjective experience

No obvious symptoms

Feeling normal ≠ no physiological change

≠

WHAT MAY BE CHANGING

Metabolic regulation	e.g. glucose changes
Blood-pressure changes	may be symptomless
Chronic inflammation	potential long-term risks

CDC / PREDIABETES	WHO / HYPERTENSION	NCI / CHRONIC INFLAMMATION
Can remain symptomless for years.	Often has no noticeable symptoms.	May increase some cancer risks.

RESEARCH INSIGHT

Not all bodily changes can be felt as they happen.

Examples illustrate possible risks—not a fixed disease pathway or individual prediction.

Long-term Evidence

Long-term dietary patterns are associated with health outcomes decades later. Rather than examining the effect of a single meal, longitudinal research reveals how sustained eating habits relate to healthy aging over time.

2025 / NATURE MEDICINE · LONGITUDINAL STUDY

105,015	30 years	1.86
PARTICIPANTS US adults; aged 39–69 at baseline	MAX. FOLLOW-UP Long-term tracking of dietary patterns	ADJUSTED ODDS RATIO Highest vs lowest AHEI adherence

Higher dietary adherence was associated with greater odds of healthy aging.

ADJUSTED ODDS RATIO / HIGHEST VS LOWEST AHEI



95% CI: 1.71–2.01 · AHEI quintiles

Association, not a personal aging prediction

Highest vs lowest dietary-pattern adherence → healthy aging association

HEALTHY AGING DEFINITION

At age 70 or older, free from major chronic diseases and maintaining cognitive, physical and mental health.

Source: Tessier et al., Nature Medicine (2025) · Observational study

Different Body Systems

Different body systems reflect distinct dietary and physiological pathways. One overall score cannot capture all their changes.



BRAIN / COGNITION

Dietary Patterns → Cognitive Health

FOCUS / Cognitive Function



CARDIOVASCULAR

Sodium Intake → Blood Pressure

FOCUS / Vascular Condition



LIVER / METABOLISM

Dietary Patterns → Liver Fat

FOCUS / Metabolic Health



MUSCLE & BONE

Protein + Activity → Tissue Maintenance

FOCUS / Muscle & Bone Condition

Organ visuals illustrate mechanisms, not measured personal damage.

Summary

01 NUTRITIONAL INTERPRETATION

Familiar foods have measurable nutrients, but quantities need meaningful reference points.

02 LONG-TERM IMPLICATIONS

Dietary habits develop through repetition, while their associations with health may unfold over many years.

03 MULTIDIMENSIONAL AGING

Different body systems undergo distinct changes, which cannot be fully described by a single health indicator.

RESEARCH SYNTHESIS

Everyday nutrient intake can be measured, while its long-term, cumulative and system-specific health relationships remain difficult to perceive. This gap motivates an interactive experience for exploration and comparison.

Insight

Based on research into young adults' eating behaviors. long-term dietary patterns, and the characteristics of aging, three perceptual gaps were identified.



01 / ABSTRACT INTAKE

Food is tangible. Nutrient data isn't.

Familiar foods are easy to recognize, but their nutrient quantities and daily reference values remain abstract and difficult to interpret.

Familiar Foods | Abstract Numbers

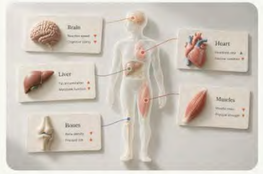


02 / INVISIBLE PROGRESSION

Some changes develop before they can be felt.

Some metabolic and cardiovascular changes may develop without noticeable symptoms, making long-term health risks difficult to perceive through everyday experience.

Hidden Changes | Delayed Awareness



03 / BODY RESPONSES

One number cannot describe the body.

Diet-related health changes involve different physiological processes across body systems, which cannot be fully represented by a single health score.

Multiple Systems | Distinct Responses

Summary

Three perceptual gaps emerge: abstract nutrient data, hidden bodily progression and differentiated system responses.

Design Opportunity

How might we make the gradual, often invisible relationships between dietary habits and bodily health tangible, observable and comparable?

01 / TANGIBLE INTAKE

Use familiar foods as tangible inputs, connecting everyday dietary choices with measurable nutrient quantities.

Cola | Chips | Chicken | Bear | Vegetables | Milk

CHOOSE · COMBINE



02 / EXPERIENTIAL TIME

Compress long-term progression into simulated time, allowing gradual bodily changes to be observed, adjusted and compared.



PROGRESS · COMPARE

03 / VISIBLE BODY RESPONSE

Translate simulated body-system responses into distinct visual feedback, revealing differences beyond a single health score.

Heart	Brain	Liver	Muscle & Bone
Heartbeat Rate	Reaction Speed	Fat Accumulation	Muscle Mass
Vessel Condition	Cognitive Clarity	Tissue Appearance	Bone Density

VISUALIZE · DIFFERENTIATE

Ideation

Six concepts explored tangible inputs, representations of time and visual metaphors for bodily change. The selected direction combines physical food cards, accelerated simulation and differentiated organ feedback.



Clinical Shell

Explores a medical-device-inspired form for presenting bodily feedback.



Walking Distance

Uses movement along a path to represent elapsed time.



Time as Motion

Represents passing years through a moving physical marker.



Physical Intake

Uses familiar food cards as tangible dietary inputs.



Living Body

Explores living forms as a metaphor for gradual bodily change.



Life Journey

Imagines dietary choices as branching life trajectories.

CONCEPT SELECTION — Selected approach from the six explorations

Physical Food Cards → Accelerated Time → Organ-specific Feedback

Design Goals

Make Dietary Input Tangible

Connect everyday food choices with measurable nutrient inputs through tangible food cards.

Make Hidden Progression Observable

Use accelerated simulations to make gradual, otherwise invisible bodily changes observable and comparable.

Reveal Differentiated Body Responses

Visualize distinct responses across four body systems rather than relying on a single health score.

Support Active Exploration and Reflection

Encourage dietary exploration, outcome comparison and reflection without prescribing a correct diet.

Final Concept

Dietary Traces is an interactive installation that transforms everyday food choices into simulated long-term body changes. Through tangible input, accelerated time and dynamic organ visualization, it helps young adults explore the often-invisible relationship between dietary habits and future health.

Physical Food Cards

Each card represents a recurring dietary input maintained over the simulated lifespan, rather than the immediate effect of a single meal.



Body System View

Four body systems reveal distinct responses to simulated dietary and exercise habits.



DIETARY TRACES

Tangible Output

A printed record captures dietary choices, changing conditions, and final results.



Simulation Engine

Turn dietary choices into nutritional and activity variables.

SUGAR · SALT · FAT · ALCOHOL · FIBRE · PROTEIN

Dynamic Visualization

An accelerated timeline makes gradual changes across the body observable.



Experience System

01 Physical Input	02 Simulation Engine	03 Dynamic Visualization	04 Tangible Output
Food cards establish a daily pattern.	Diet and exercise update body states.	Aging reveals changing system responses.	A printed record summarizes results.

Interaction Flow

FROM INPUT TO REFLECTION

START / SET HABITS

USER INTERACTION	SYSTEM RESPONSE
01 INPUT **Choose food cards** Set exercise level	**Recognize inputs** NFC card IDs → nutrient data
02 SIMULATE **Start simulation** Control simulated time	**Initialize model** Baseline · diet · activity
03 OBSERVE **Explore organ views** Switch four body systems	**Render body responses** Changing visuals and trajectories
04 ADJUST **Adjust habits midway** Change food or exercise	**Update future trajectories** Past simulated exposure remains
05 REFLECT **Review & print takeaway** Compare simulated changes	**Generate experience record** Inputs · progression · takeaway

COMPARE & REFLECT

Rather than judging choices as right or wrong, participants can adjust their habits, observe evolving body responses, and explore how past choices shape future simulated trajectories.

Data Translation

Food cards become recurring nutrient inputs, which the model translates into changing conditions across four body systems over time. This input-to-feedback process also guides the installation's physical controls, sensing, and output design.

1 Food to Nutrient Input

Each card represents a recurring daily intake, translated into six quantifiable dietary factors.



2 Nutrient to Body Systems

Multiple nutrients influence multiple body systems; these links are a qualitative model diagram.



Selected qualitative pathways; numerical influence coefficients remain in the simulation specification.

1 Example · Cola Card (Sugar)

A card adds to the model's daily baseline intake.

35 g	+	12 g	=	47 g
COLA CARD		BASELINE		DAILY TOTAL

Normalized to the model's 50 g reference **0.94**

Relative input level, not a health-risk percentage.

3 Dynamic State Model

Each organ's target state combines age-related change, dietary burdens, protective intake, exercise, and accumulated effects.

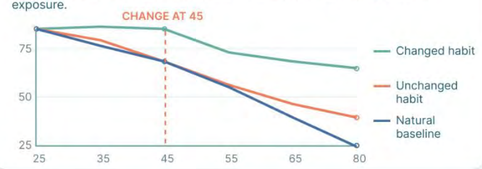
$$U_m = B_m - D_m + P_m + G_m - S_m$$

- B_m Aging baseline
- D_m Dietary burden
- P_m Protective intake
- G_m Exercise effect
- S_m Accumulated burden

Target state ≠ instant state **Past habits still matter**

4 Habit to Aging Trajectory

Changing habits alters the subsequent trajectory without erasing previous exposure.



Hardware Design

Sketch

1 Clinical shell

The console and screen are integrated, similar to a medical scanner.



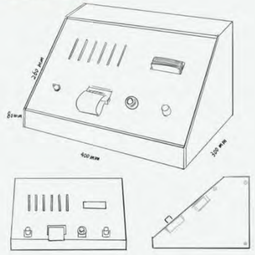
2 Monitor Plan

A separate display module positioned above the control interface for clear visualization and comfortable viewing.



3 Enclosure Sketch

Explores the overall form,dimensions and key component layout of the tabletop unit.



4 Card Sensing Solution

Consider forms like flat placement card insertion, buttons, etc.



Component & Operation

EC11 rotary encoder



Controls simulation time through rotation, while its integrated button starts or pauses the simulation.

INPUT Rotate / Press › PROCESS Set Time › FEEDBACK Time Moves

360-degree EC11 rotary encoder



Adjusts exercise intensity from 0% to 100%, modifying the activity variable to simulate different lifestyle conditions.

INPUT Rotate Knob › PROCESS Set 0–100% › FEEDBACK Body Updates

Dual-axis button joystick



Switches between four body systems, allowing participants to explore and compare their simulated responses.

INPUT Move Stick › PROCESS Select Organ › FEEDBACK View Switches

NFC module RC522



Six NFC readers identify inserted food cards and convert their IDs into nutritional data, updating the dietary combination as cards change.

INPUT Insert Card › PROCESS Map Card ID › FEEDBACK Diet Updates

TCA9548 Multi-Channel Expander



Expands communication channels, enabling the main controller to manage multiple RC522 card readers independently.

INPUT RC522 Data › PROCESS Route Data › FEEDBACK Main Board

Thermal printer

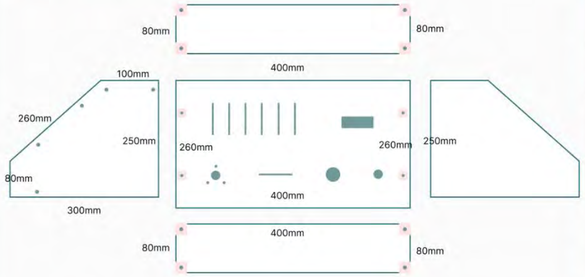


Prints a summary of dietary choices and simulation results, allowing participants to take away a physical record of their experience.

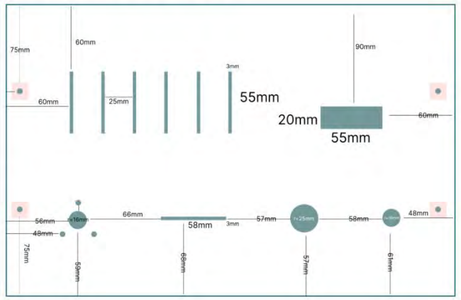
INPUT Finish › PROCESS Summarize › FEEDBACK Receipt Out

Panel Design

5 Panel Sizes:



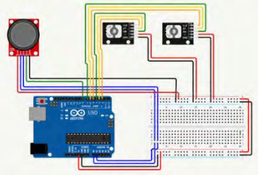
Panel Cutout Design:



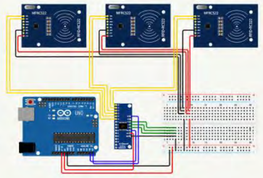
Circuit Design

The diagram shows the wiring logic; the specific model and wiring method should be based on the actual situation.

Joystick EC11 rotary encoder



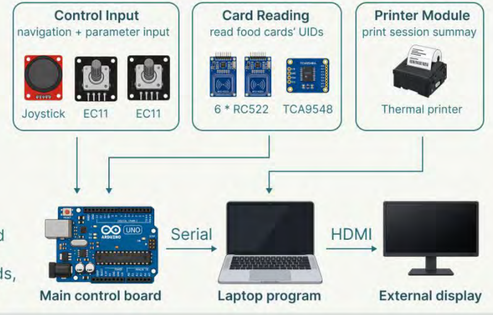
NFC module RC522



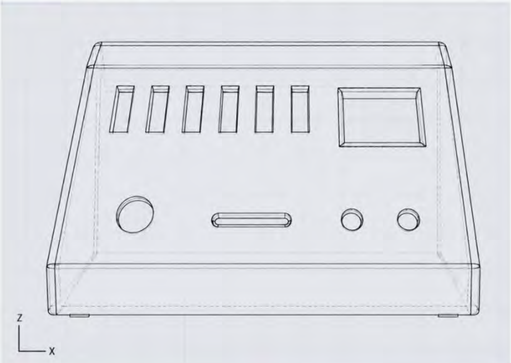
The rotary potentio meter and joystick act as the control unit and are managed separately by a control board, providing signal input for the final program.

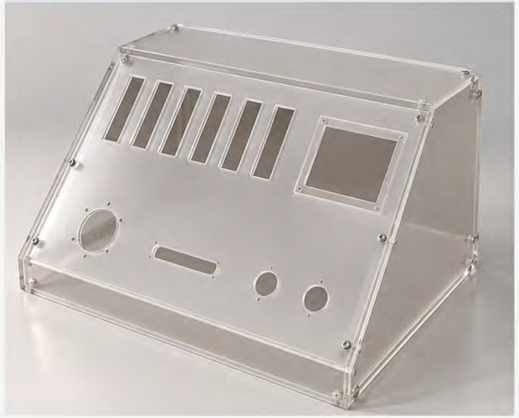
Multiple sensor modules are multiplexed and expanded, controlled by a control board. They act as a sensing queue for inserting food cards, sending signals to the final program.

Physical input, card sensing and print output are coordinated by the main program and translated into real-time visualization.



Modeling

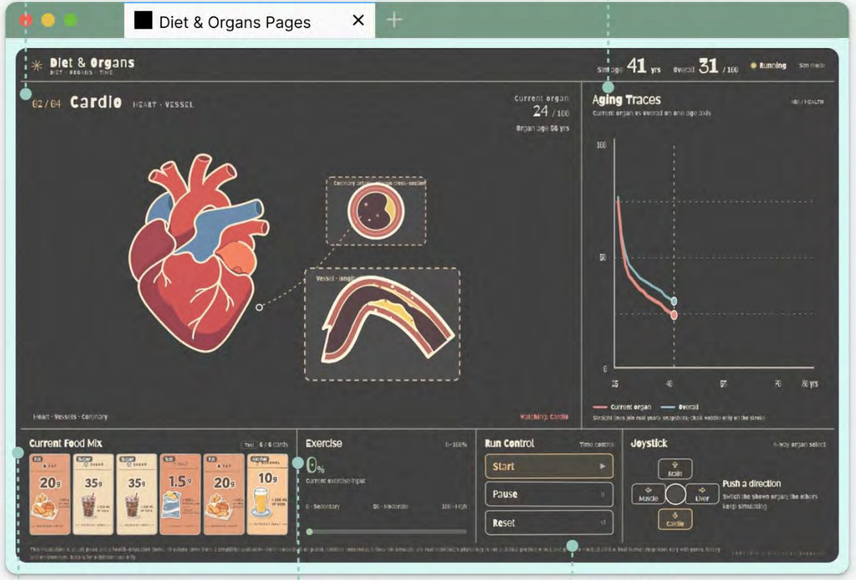




Interface

Organ Imaging
- A visual display of organ health that turns the effects of diet into intuitive images.
- Organ condition updates in real time with changes in age, diet, and exercise habits.

Aging Traces
- Real-time comparison chart of ideal organ age versus actual age.



Current Food Mix
- The food combinations inserted in the display device in real time can change instantly as the user switches them.
- There are two ways to show them: with pictures or text.

Exercise Intensity
- Supports adjustments from 0% to 100%, reflecting the user's daily exercise habits.
- Increasing exercise intensity can offset some of the intake of sugar and fat.

Run Control
- Start to see the changes, pause to switch decks, reset to return to the beginning.

Joystick
- Switch the organs being shown on the screen.

Organ Views

Each organ requires a distinct visual language to communicate simulated changes. Recognizable forms and carefully selected visual features make different bodily responses easier to identify and compare.

Style Definition — Visual exploration · four approaches



Hand-drawn tech



Data cable sketch



Minimalist life grid



Chalkboard style

DESIGN DIRECTION

Recognizable, not overly realistic

From detailed anatomical rendering to simplified line-based graphics. Clear contours and restrained textures make changes easier to compare.

Dynamic Changes

01 Brain & Nervous System



Healthy Worse

Signal rhythm, connectivity, and image clarity reflect the simulated cognitive state of the brain.

VISIBLE FEATURES — Reaction rhythm · Clarity

02 Musculoskeletal System

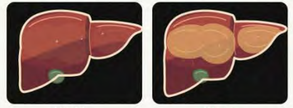


Healthy Worse

Muscle size, structure, contraction speed, and bone density reflect the health of this system.

VISIBLE FEATURES — Muscle mass · Bone density

03 Liver & Metabolic System



Healthy Worse

Changes in color, tissue texture and fat distribution suggest different simulated metabolic states.

VISIBLE FEATURES — Fat accumulation · Tissue texture

04 Cardiovascular System



Healthy Worse

The state of the cardiovascular system is shown through heart rate, the strength of the beat, and the size of any visible plaque in the blood vessels.

VISIBLE FEATURES — Heart rhythm · Vessel condition

Receipt

Body Receipt
Diet & Body Simulation

1. Experience Record

EXP ID	EXP-2026-0602-001
Time	2026-06-02 23:19
Span	Age 25-80 55Years

2. Your Diet

Daily Habit Combos
Combo1

Card	Qty	Intake	Goal
Sugar	*2	82g/day	164%
Salt	*0	5g/day	100%
Fat	*2	65g/day	295%
Alcohol	*2	20g/day	-
Fiber	*0	10g/day	20%
Protein	*0	10g/day	80%

3. Body Response

Organ age at the end
[Heart] 66Years
Peer norm 80----14years
[Brain] 61Years
Peer norm 80----19years
[Liver] 68Years
Peer norm 80----12years
[Bone] 72Years
Peer norm 80---- 8years

Avg Organ Age	86.3years
Vs Actual Age	+6.3years

4. Your Trace

One Line Summary

You and your body broke even.

Cards Design

Each card connects a familiar food with a measurable nutrient input. Its two sides separate dietary information from recognizable food imagery.

Front
Highlight the dietary elements and their intake amounts to help users make connections.

Back
Use easily recognizable food illustrations to make card selection more intuitive and friendly.





Prototyping

Model Making



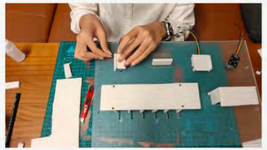
Model Shell Assembly
Assemble and secure the custom acrylic panels using nuts, screws, and hot glue.



Part distribution
Determine what kind of internal structure to make based on testing the part placement according to the hole positions.



Make the internal structure
Making structures inside the shell that matches the size of the part to support placing the part.

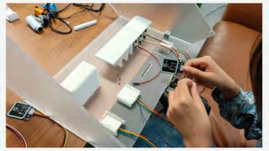


Internal assembly
After making the internal parts, use hot glue and thumbtacks to attach the KT board inside the casing.

Technical Implementation



Circuit board placement
Confirm how many circuit boards are needed and where the boards can be placed.



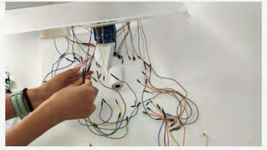
Parts Assembly
Plug a matching DuPont wire into each module and lengthen the wires as needed.



Print Module
Remove the circuit board from the printer to make installation easier.



Printer location
Test to make sure the printer can feed paper normally and won't get stuck on the panel.



Wiring design
Plan which components connect to which circuit board.



Component Wiring
Use three circuit boards to power up and test all the components.



Appearance enhancement
Use metal letters to label the purpose of each part on the acrylic panel.



Cards Making
Stick the prepared content of the card on both sides of the NFC sensing card.

Hardware Coding

The hardware code manages NFC card recognition and rotary encoder inputs, converting physical interactions into digital control signals.

Multi-sensor communication polling

```
void pollReaderSlot(Reader *r, unsigned long now) {
  if (!r->present) return;                    // nothing to poll when absent
  if (now - r->lastPollMs < POLL_MS) return;  // not this slot's turn yet
  r->lastPollMs = now;

  muxSelect(r->mux);                          // route I2C to this reader
  uint8_t ver = r->nfc.getFirmwareVersion();  // cheap "is it still there?" probe
  delay(STABLE_WARM_MS);

  byte uid[10], uidLen = sizeof(uid);
  bool read = ver && r->nfc.readPassiveTargetID(PN532_MIFARE_ISO14443A, uid, &uidLen, 50);

  if (read) {                                 // a card is on the antenna
    r->lastSeenMs = now;
    if (!r->present) {                        // absent -> present: debounce
      uint32_t id = uidToId(uid, uidLen);
      r->nfc.setPassiveActivationRetries(0);  // don't block the next reader
      r->nfc.SAMConfig();
      if (id == r->candidateId) {
        if (++r->candCount >= DEBOUNCE) {     // seen enough times in a row
          r->present = true;
          r->uidLen = uidLen;
          memcpy(r->uid, uid, uidLen);
          r->candCount = 0;
          if (snapshotMode) {
            printReaderDiag(r);
            reportSnapshot();
          }
        }
      } else {                                // first sighting of a new card
        r->candidateId = id;
        r->candCount = 1;
      }
    }
  } else if (r->present && now - r->lastSeenMs > PRESENT_MS) {
    r->present = false;                       // absent: clear immediately
    r->candidateId = 0;
    r->candCount = 0;
    Serial.println(F("-,removed"));
  }
}
```

Multi-channel sensor communication

```
void muxSelect(byte ch) {
  Wire.beginTransmission(MUX_ADDR);
  Wire.write((uint8_t)(1 << ch));
  Wire.endTransmission();
}

/* Scan the currently active channel bus; return the first ACKing address
 * (0 = no device). The mux itself is skipped: 0x70 sits on the main bus and
 * never disconnects with a channel -- scanning it would echo the register
 * address we just wrote (VersionReg=0x37 -> fake bad-version). */
byte scanChannel() {
  for (byte a = 1; a < 127; a++) {
    if (a == MUX_ADDR) continue;
    Wire.beginTransmission(a);
    if (Wire.endTransmission() == 0) return a;
  }
  return 0;
}

/* UID byte array -> first 4 bytes, big-endian, packed into uint32_t */
uint32_t uidToId(byte *uidByte, byte size) {
  uint32_t v = 0;
  byte n = size < 4 ? size : 4;
  for (byte i = 0; i < n; i++) v = (v << 8) | uidByte[i];
  return v;
}
```

Rotary Encoder - Motion Control Code

```
/* ---------- Encoder: polled 4-state Gray-code decoding ----------
 * Count one step only on a CLK-only falling edge, then read DT for direction:
 * DT high = clockwise +1, DT low = counter-clockwise -1; simultaneous
 * / two-bit toggles are rejected as glitches. If rotation is reversed on the
 * physical build, swap the final +1/-1 (field-tunable). */
int pollEncoderStep(Encoder *e) {
  byte cur = (byte)((digitalRead(e->pinClk) << 1) | digitalRead(e->pinDt));
  if (cur == e->code) return 0;               // no state change
  byte diff = cur ^ e->code;
  e->code = cur;
  if (diff != 0b10) return 0;                 // valid edge only when CLK alone toggles
  if (cur & 0b10) return 0;                   // rising edge: ignored
  return (cur & 0b01) ? -1 : 1;               // falling edge decides direction
```

Rotary Encoder - Switch Control

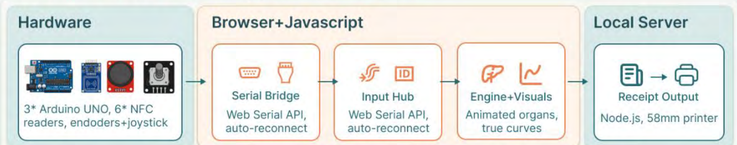
```
#define DETENTS_PER_TURN 19          // 编码器一圈脉数（2026-09-28 实测）
int timePos       = 0;               // 累计档位 0..18（mod 19），上电后非置0
int timeLastOpt   = -1;              // 上次上报的档位（检查到时间 timeSelect）
const char *const TIME_VALUES[3] = { "reset", "pause", "start" };

/* Detent -> option sector: 0=reset 1=pause 2=start (boundaries 0/5/9 as tuned) */
int timeOption(int p) {
  if (p >= 16 || p <= 2) return 0;   // reset zone (incl. detent 0)
  if (p <= 8)            return 1;   // pause zone (incl. detent 5)
  return 2;                          // start zone (incl. detent 9)
}

byte joyIdx = 0;                     // 摇杆状态: 0=none 1=up 2=down 3=left 4=right（仅后三项）
```

Software Coding

Web software structure for the dietary traces interactive system.



Three Arduino boards send card and control events via Web Serial to a unified Input Hub. A diet-organ dynamics engine drivers hand-drawn SVG organs and real-data curves, while a local Node.js bridge handles thermal receipt printing.

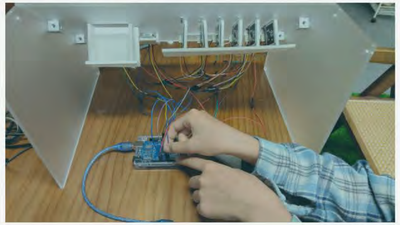
JavaScript ES5 · Web Serial API · SVG · WebSocket

AI-assisted Development
AI helped making architecture, implementations and debugging tools.Human-led assembly, field testing and design decisions.

GitHub Repository:
https://github.com/ZRYiiii-web/Dietary-Traces
https://zryiiii-web.github.io/Dietary-Traces/

Testing

NFC Sensing Test



Get the ID of each NFC card and enter the food information, making sure it's recognized correctly.

Check multiple cards detecting at the same time, and look for any interference between them.

Part Testing



Check the NFC sensing accuracy and troubleshoot any mutual interference.

Wiring Test



Check if the wiring method works and make sure there's enough power.

On-site test





Check the on-site equipment placement and wiring layout, connect the devices, laptop, and monitor, and make sure the system is running properly.

✱ Outcome



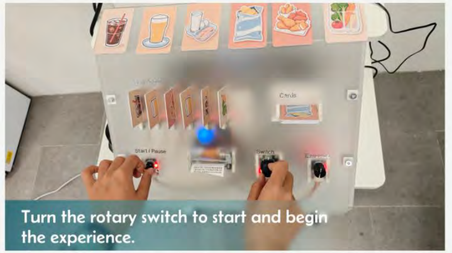
✱ Scenario



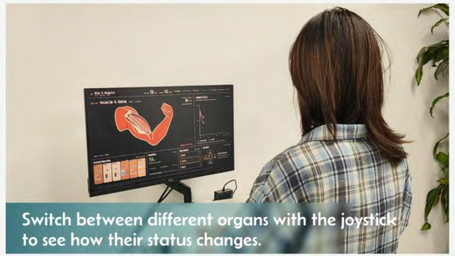
Browse all the food cards and choose the combination of eating habits you want to try.



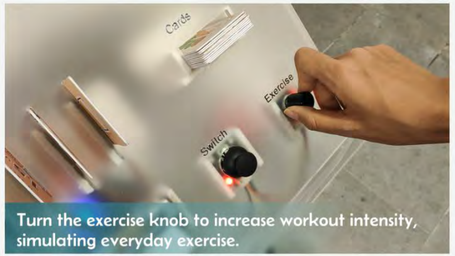
Insert the selected card combo into the device slot.



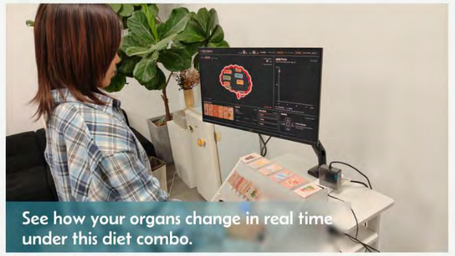
Turn the rotary switch to start and begin the experience.



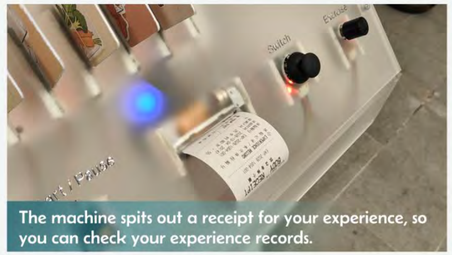
Switch between different organs with the joystick to see how their status changes.



Turn the exercise knob to increase workout intensity, simulating everyday exercise.



See how your organs change in real time under this diet combo.



The machine spits out a receipt for your experience, so you can check your experience records.

Video Link : https://youtu.be/OPF0A04r6mc

✱ Reflection

This project taught me how to translate abstract data into tangible interactions and visual experiences. While I used simplified simulations to communicate bodily changes, future iterations could explore more scientifically grounded models, personalized dietary inputs, and richer physical feedback. I also realized that user testing could help refine the interaction and make the experience more intuitive and engaging.